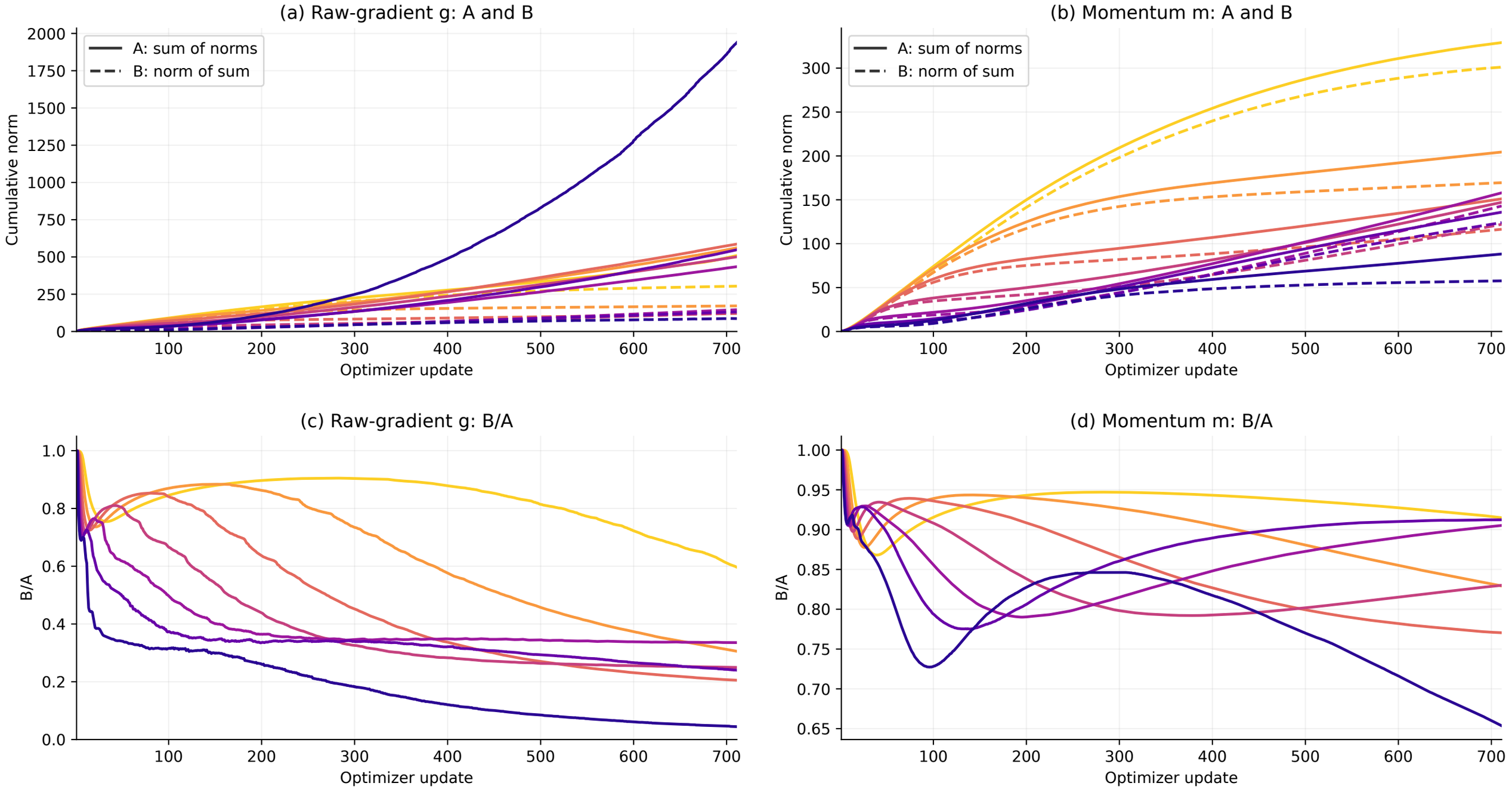
Pythia-14M: cumulative raw-gradient and momentum geometry across learning rates

Peak LR: light yellow to dark purple as LR increases; S = Spark, R = RTX4090

0.0001 (S) · 0.000215443 (R) · 0.000464159 (R) · 0.001 (S) · 0.00215443 (R) · 0.00464159 (R) · 0.01 (S)



Colors identify LR using reversed plasma, uniformly sampled in log10(LR); higher LR is darker.
Top: solid A = sum of vector norms, dashed B = norm of vector sum. Bottom: B/A; all curves unsmoothed, linear Y.
g is raw before clipping; m is stored uncorrected Adam exp_avg from clipped gradients, before preconditioning.
Display: updates 1-711; partial update 712 retained in results. Seed 1234; hardware differs across S/R; LR scales decay.